

DDoS Incident Report

Report Period	Sep 26, 2026 22:16 - Sep 26, 2026 22:21
Generated At	2026-09-26 22:21:59
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

Metric	Value	Traffic Summary	Count
Total Threats Mitigated	28	Total Flows Observed	15330
High Priority	13	Total Packets Analyzed	8019908468
Low Priority	15	Malicious Packets	7999458959
Manual Releases	0	True Negatives Passed	10375
Manual Blocks	0		
ML Metrics	Value	Attack Vector	Count
False Positives	96	ICMP Flood	0
FP Rate	0.75%	SYN Flood	0
IF Precision	98.45%	UDP Flood	22
IF Recall	93.94%	Uncertain	6
IF F1-Score	96.14%		
IF Accuracy	97.47%	Action	Count
		Quarantined	5
		Time Ban	3
		Blackhole	0
		Released	8

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	98.45%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.94%	Share of actual attacks successfully flagged
F1-Score	96.14%	Balanced measure combining Precision and Recall
Accuracy	97.47%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.75%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	6.06%	Actual attacks that went undetected
True Positive Rate (TPR)	93.94%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.25%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4818	FN 311
Actual: Normal	FP 76	TN 10064

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.83%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.92%	Balanced measure combining Precision and Recall
Accuracy	99.84%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	10.53%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	89.47%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	4818

2c. Hybrid Cascaded

Metric	Value	Description
Precision	98.45%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.94%	Share of actual attacks successfully flagged
F1-Score	96.14%	Balanced measure combining Precision and Recall
Accuracy	97.47%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.75%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	6.06%	Actual attacks that went undetected
True Positive Rate (TPR)	93.94%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.25%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4818	FN 311
Actual: Normal	FP 76	TN 10064

2d. Response Latency

Metric	Value	Description
Detection Time	537.1 ms	Average time taken to detect an attack
Mitigation Response Time	11.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.00%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	3.88%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.77%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.20	5	4	2026-09-26 22:18:46	2026-09-26 22:21:26
10.0.0.22	5	4	2026-09-26 22:18:47	2026-09-26 22:21:27
10.0.0.21	5	3	2026-09-26 22:18:47	2026-09-26 22:21:31
10.0.0.8	2	2	2026-09-26 22:19:05	2026-09-26 22:19:57
10.0.0.9	2	2	2026-09-26 22:18:06	2026-09-26 22:19:15
10.0.0.1	2	1	2026-09-26 22:19:10	2026-09-26 22:19:42
10.0.0.10	2	1	2026-09-26 22:17:59	2026-09-26 22:18:30
10.0.0.5	2	1	2026-09-26 22:17:52	2026-09-26 22:18:14

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 22:18:03	10.0.0.5	DDoS	Uncertain	58.9%	Low	Released
2026-09-26 22:18:06	10.0.0.9	DDoS	UDP Flood	92.4%	High	Time Ban (30s)
2026-09-26 22:18:09	10.0.0.10	DDoS	Uncertain	68.7%	Low	Released
2026-09-26 22:18:36	10.0.0.9	DDoS	UDP Flood	92.4%	High	Released
2026-09-26 22:18:46	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 22:18:47	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 22:18:47	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 22:18:56	10.0.0.20	DDoS	UDP Flood	75.6%	High	Time Ban (30s)
2026-09-26 22:18:57	10.0.0.20	DDoS	UDP Flood	75.6%	High	Time Ban
2026-09-26 22:18:57	10.0.0.21	DDoS	UDP Flood	86.4%	High	Time Ban (30s)
2026-09-26 22:18:57	10.0.0.22	DDoS	UDP Flood	86.1%	High	Time Ban (30s)
2026-09-26 22:18:57	10.0.0.22	DDoS	UDP Flood	86.1%	Critical	Time Ban
2026-09-26 22:18:57	10.0.0.21	DDoS	UDP Flood	86.4%	Critical	Time Ban
2026-09-26 22:19:05	10.0.0.9	DDoS	UDP Flood	92.4%	High	Quarantined
2026-09-26 22:19:05	10.0.0.8	DDoS	UDP Flood	91.6%	High	Time Ban (30s)
2026-09-26 22:19:20	10.0.0.1	DDoS	Uncertain	58.9%	Low	Released
2026-09-26 22:19:26	10.0.0.20	DDoS	UDP Flood	85.5%	High	Released
2026-09-26 22:19:27	10.0.0.21	DDoS	UDP Flood	86.4%	High	Released
2026-09-26 22:19:27	10.0.0.22	DDoS	UDP Flood	86.1%	High	Released
2026-09-26 22:19:35	10.0.0.8	DDoS	UDP Flood	91.9%	High	Released
2026-09-26 22:19:37	10.0.0.20	DDoS	UDP Flood	85.5%	Critical	Time Ban (1m)
2026-09-26 22:19:39	10.0.0.22	DDoS	UDP Flood	86.1%	Critical	Time Ban (1m)
2026-09-26 22:19:41	10.0.0.21	DDoS	UDP Flood	86.4%	Critical	Time Ban (1m)
2026-09-26 22:19:46	10.0.0.8	DDoS	UDP Flood	91.9%	High	Quarantined
2026-09-26 22:20:38	10.0.0.20	DDoS	UDP Flood	85.5%	Critical	Time Ban (2m)
2026-09-26 22:20:40	10.0.0.22	DDoS	UDP Flood	88.1%	Critical	Time Ban (2m)
2026-09-26 22:21:27	10.0.0.22	DDoS	UDP Flood	77.1%	Critical	Time Ban (5m)
2026-09-26 22:21:31	10.0.0.21	DDoS	UDP Flood	79.2%	Critical	Time Ban (5m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.6449	79.2%	Critical	Phase 2	1	10s	2026-09-26 22:21:31	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6298	77.1%	Critical	Phase 2	1	10s	2026-09-26 22:21:27	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6448	75.7%	Critical	Phase 2	1	10s	2026-09-26 22:21:26	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6072	90.1%	High	Phase 2	1	1m 12s	2026-09-26 22:20:41	Ban Expired
10.0.0.22	UDP Flood	0.6330	88.1%	Critical	Phase 2	4	1m 11s	2026-09-26 22:20:40	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6143	85.5%	Critical	Phase 2	4	1m 11s	2026-09-26 22:20:38	Attack persisted - escalating to time ban
10.0.0.8	UDP Flood	0.6328	91.9%	High	Phase 1	2	11s	2026-09-26 22:19:57	Attack Stopped
10.0.0.1	Uncertain	0.6044	58.9%	Low	Phase 1	1	11s	2026-09-26 22:19:42	Attack Stopped
10.0.0.21	UDP Flood	0.6466	86.4%	Critical	Phase 2	3	12s	2026-09-26 22:19:41	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6386	86.1%	Critical	Phase 2	3	10s	2026-09-26 22:19:39	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6349	85.5%	Critical	Phase 2	3	10s	2026-09-26 22:19:37	Attack persisted - escalating to time ban
10.0.0.8	UDP Flood	0.5688	91.9%	High	Phase 2	1	30s	2026-09-26 22:19:35	Ban Expired
10.0.0.22	UDP Flood	0.6360	86.1%	High	Phase 2	1	40s	2026-09-26 22:19:27	Ban Expired
10.0.0.21	UDP Flood	0.6418	86.4%	High	Phase 2	1	40s	2026-09-26 22:19:27	Ban Expired
10.0.0.20	UDP Flood	0.6130	85.5%	High	Phase 2	1	40s	2026-09-26 22:19:26	Ban Expired
10.0.0.1	Uncertain	0.6061	58.9%	Low	Phase 1	1	10s	2026-09-26 22:19:20	Attack Stopped
10.0.0.9	UDP Flood	0.5699	92.4%	High	Phase 1	2	10s	2026-09-26 22:19:15	Attack Stopped
10.0.0.22	UDP Flood	0.6386	86.1%	High	Phase 2	1	10s	2026-09-26 22:18:57	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6464	86.4%	High	Phase 2	1	10s	2026-09-26 22:18:57	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	75.6%	High	Phase 2	1	10s	2026-09-26 22:18:56	Attack persisted - escalating to time ban
10.0.0.9	UDP Flood	0.6095	92.4%	High	Phase 2	1	30s	2026-09-26 22:18:36	Ban Expired
10.0.0.10	Uncertain	0.6244	68.7%	Low	Phase 1	1	11s	2026-09-26 22:18:30	Attack Stopped
10.0.0.5	Uncertain	0.6042	58.9%	Low	Phase 1	1	11s	2026-09-26 22:18:14	Attack Stopped
10.0.0.10	Uncertain	0.5976	68.7%	Low	Phase 1	1	10s	2026-09-26 22:18:09	Attack Stopped
10.0.0.5	Uncertain	0.6082	58.9%	Low	Phase 1	1	11s	2026-09-26 22:18:03	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME
Network Administrator

DATE